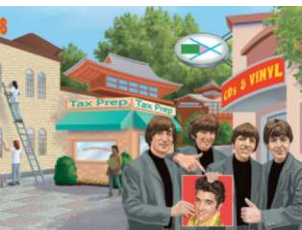


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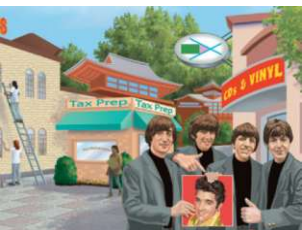
Consumers, Producers, and the Efficiency of Markets





One Principle

- Market is better way to allocate resources?
 - Maximizes the welfare?
 - Consumer welfare- Consumer surplus
 - Producers welfare- Producers surplus



Consumer Surplus

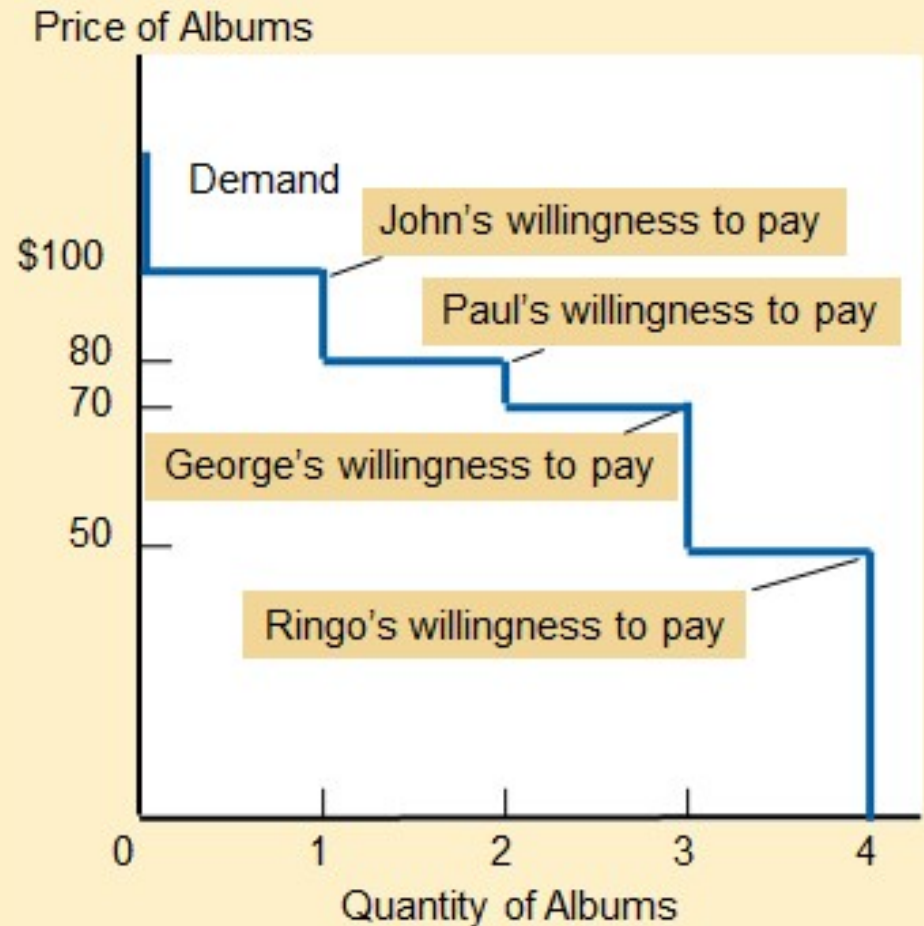
- Willingness to pay
 - Maximum amount that a buyer will pay for a good
 - How much that buyer values the good

Table 1 Four Possible Buyers' Willingness to Pay

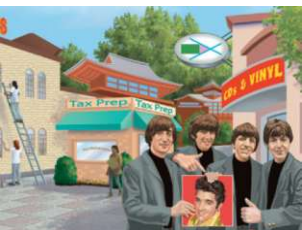
Buyer	Willingness to Pay
John	\$100
Paul	80
George	70
Ringo	50

Figure 1 The Demand Schedule and the Demand Curve

Price	Buyers	Quantity Demanded
More than \$100	None	0
\$80 to \$100	John	1
\$70 to \$80	John, Paul	2
\$50 to \$70	John, Paul, George	3
\$50 or less	John, Paul, George, Ringo	4



The table shows the demand schedule for the buyers (listed in Table 1) of the mint-condition copy of Elvis Presley's first album. The graph shows the corresponding demand curve. Note that the height of the demand curve reflects the buyers' willingness to pay.

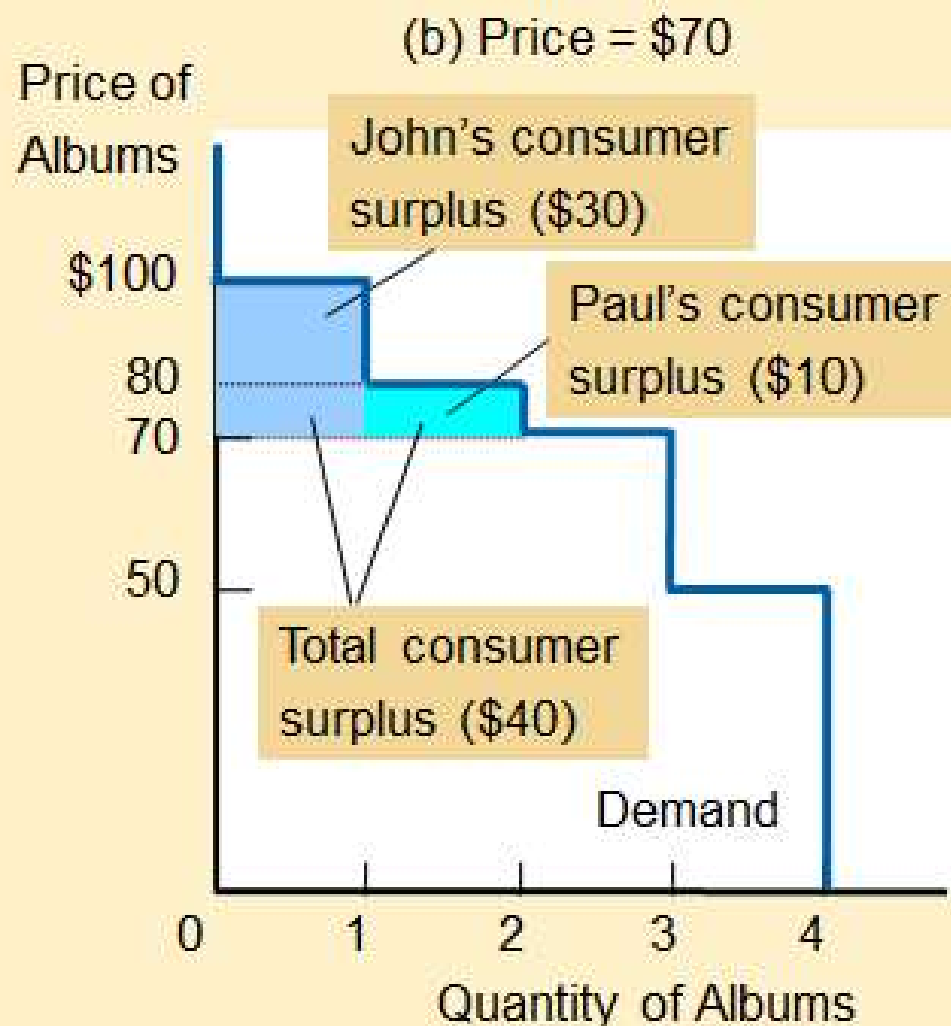
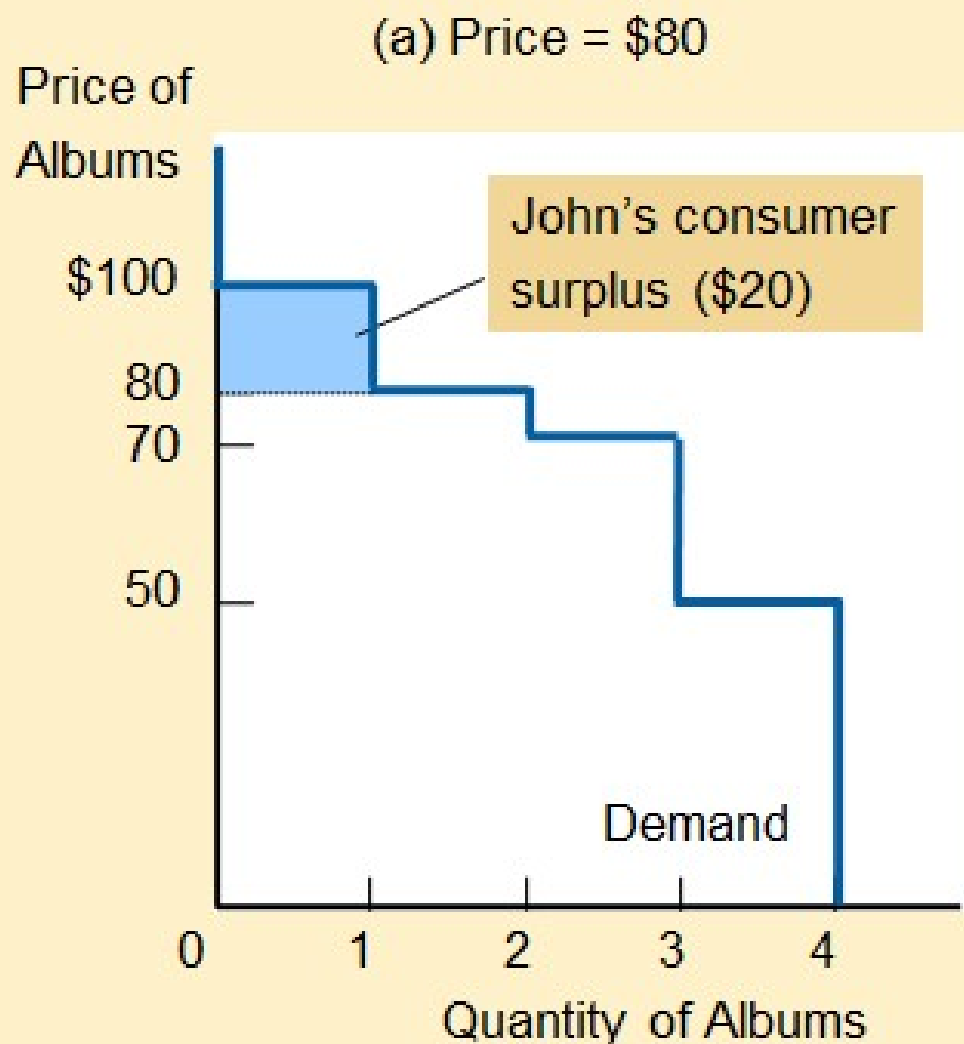


Consumer Surplus

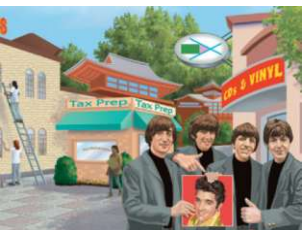
- Consumer surplus

- Amount a buyer is willing to pay for a good minus amount the buyer actually pays for it
- Willingness to pay minus price paid

Figure 2 Measuring Consumer Surplus with the Demand Curve



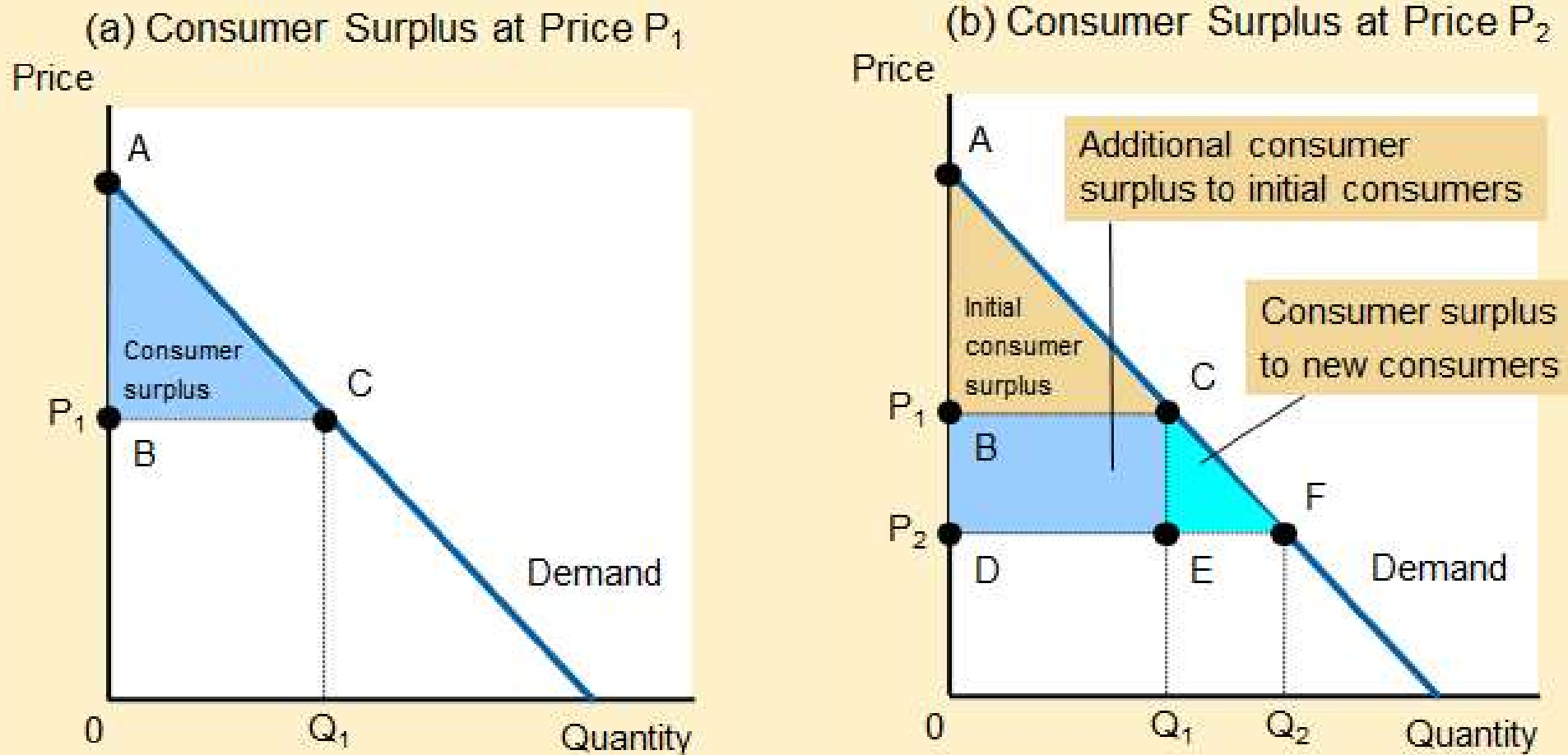
In panel (a), the price of the good is \$80 and the consumer surplus is \$20. In panel (b), the price of the good is \$70 and the consumer surplus is \$40.



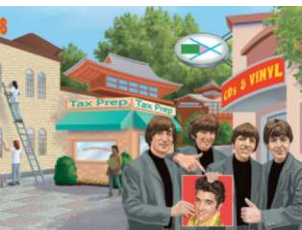
Consumer Surplus

- Consumer surplus
 - Measures the benefit buyers receive from participating in a market
- Demand schedule
 - Derived from the willingness to pay of the possible buyers
- Consumer surplus in a market
 - Area below the demand curve and above the price

Figure 3 How Price Affects Consumer Surplus

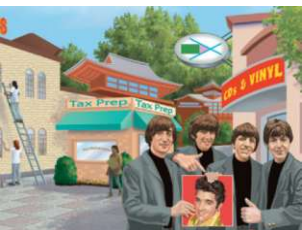


In panel (a), the price is P_1 , the quantity demanded is Q_1 , and consumer surplus equals the area of the triangle ABC. When the price falls from P_1 to P_2 , as in panel (b), the quantity demanded rises from Q_1 to Q_2 , and the consumer surplus rises to the area of the triangle ADF. The increase in consumer surplus (area BCFD) occurs in part because existing consumers now pay less (area BCED) and in part because new consumers enter the market at the lower price (area CEF).



Consumer Surplus

- A lower price raises consumer surplus
 1. Existing buyers: increase in consumer surplus
 - Buyers who were already buying the good at the higher price are better off because they now pay less
 2. New buyers enter the market: increase in consumer surplus
 - Willing to buy the good at the lower price



Producer Surplus

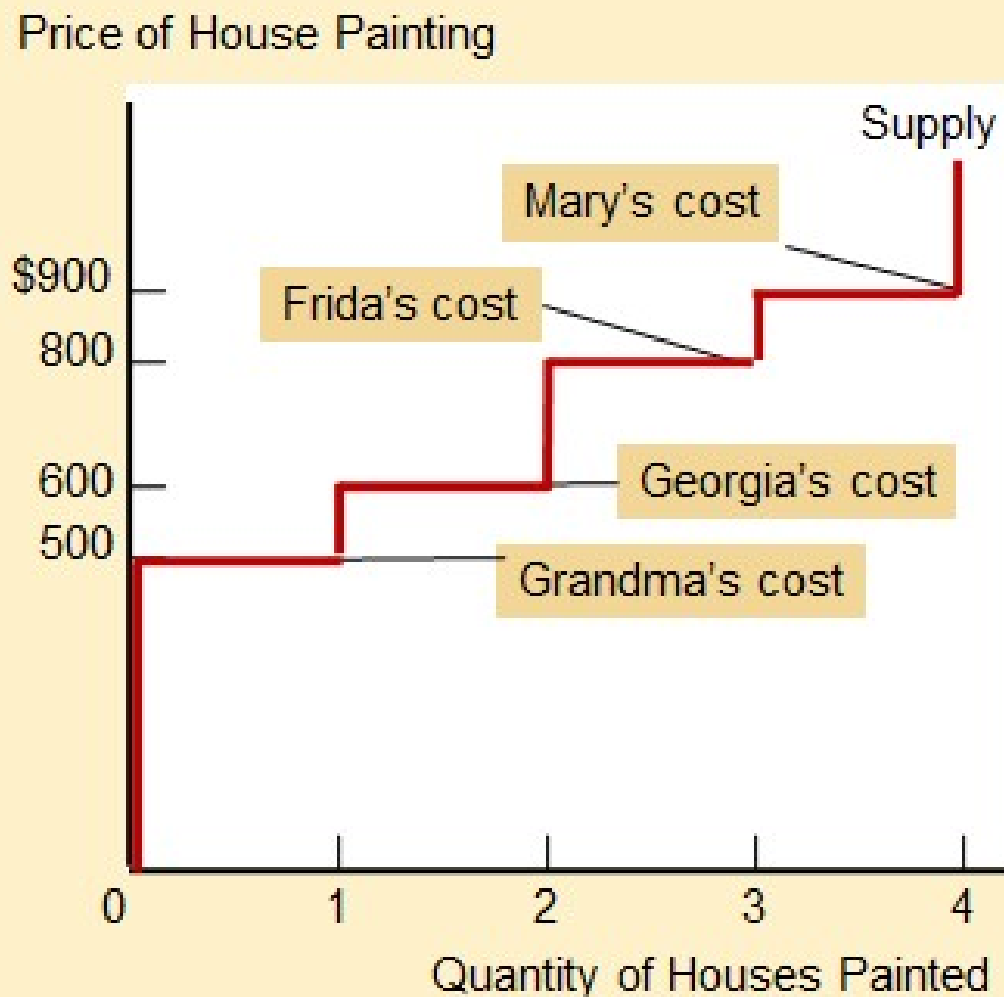
- Cost
 - Value of everything a seller must give up to produce a good
 - Measure of willingness to sell or willingness to accept

Table 2 The Costs of Four Possible Sellers

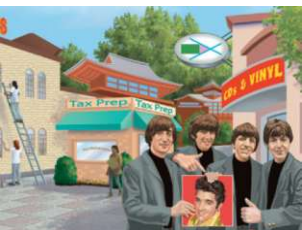
Seller	Cost
Mary	\$900
Frida	800
Georgia	600
Grandma	500

Figure 4 The Supply Schedule and the Supply Curve

Price	Sellers	Quantity Supplied
\$900 or more	Mary, Frida, Georgia, Grandma	4
\$800 or \$900	Frida, Georgia, Grandma	3
\$600 to \$800	Georgia, Grandma	2
\$500 to \$600	Grandma	1
Less than \$500	None	0

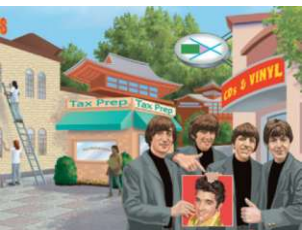


The table shows the supply schedule for the sellers in Table 2. The graph shows the corresponding supply curve. Note that the height of the supply curve reflects sellers' costs.



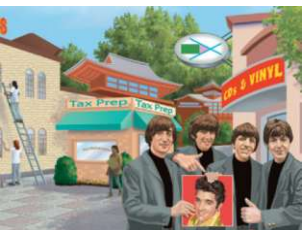
Producer Surplus

- Producer surplus
 - Amount a seller is paid for a good minus the seller's cost of providing it
 - Price received minus willingness to sell



Producer Surplus

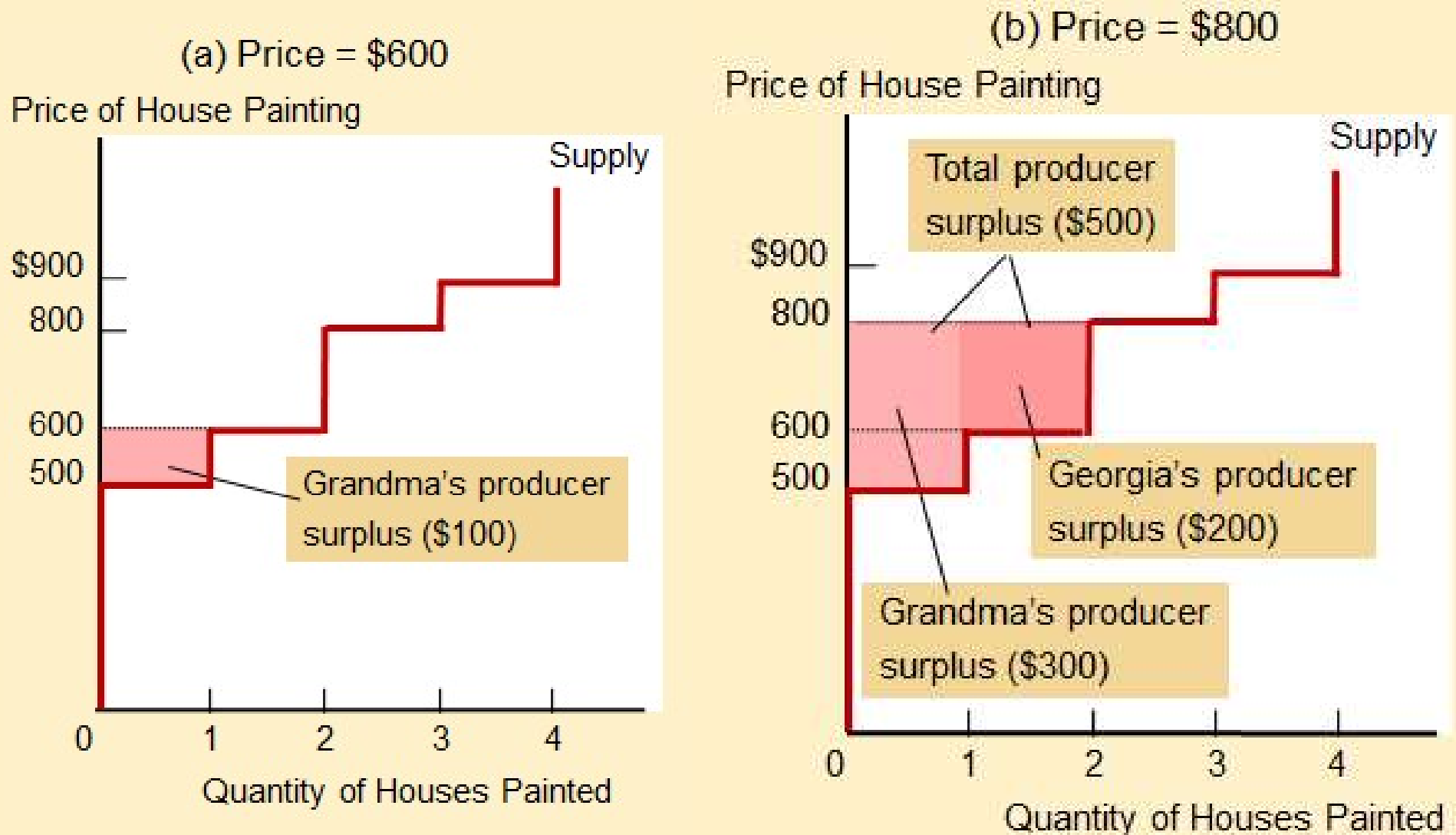
- Producer surplus
 - Closely related to the supply curve
- Supply schedule
 - Derived from the costs of the suppliers



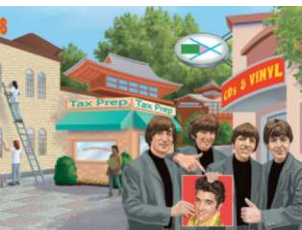
Producer Surplus

- Supply curve
 - Reflects sellers' costs
 - Used to measure producer surplus
- Producer surplus in a market
 - Area below the price and above the supply curve

Figure 5 Measuring Producer Surplus with the Supply Curve



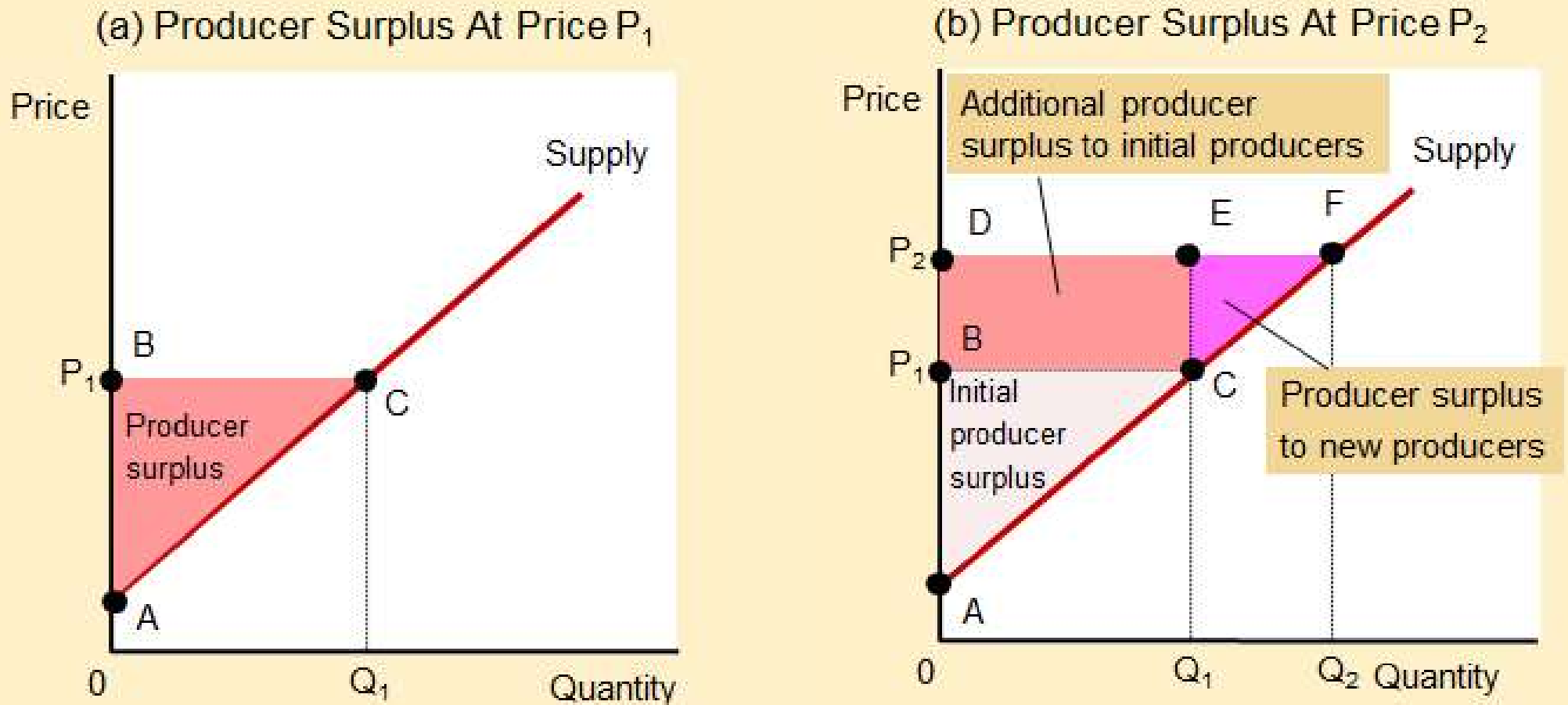
In panel (a), the price of the good is \$600, and the producer surplus is \$100. In panel (b), the price of the good is \$800, and the producer surplus is \$500.



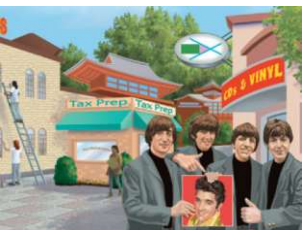
Producer Surplus

- A higher price raises producer surplus
 1. Existing sellers: increase in producer surplus
 - Sellers who were already selling the good at the lower price are better off because they now get more for what they sell
 2. New sellers enter the market: increase in producer surplus
 - Willing to produce the good at the higher price

Figure 6 How Price Affects Producer Surplus

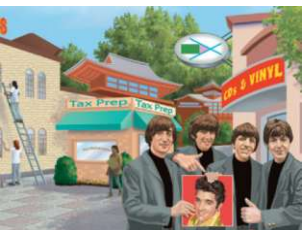


In panel (a), the price is P_1 , the quantity supplied is Q_1 , and producer surplus equals the area of the triangle ABC. When the price rises from P_1 to P_2 , as in panel (b), the quantity supplied rises from Q_1 to Q_2 , and the producer surplus rises to the area of the triangle ADF. The increase in producer surplus (area BCED) occurs in part because existing producers now receive more (area BCED) and in part because new producers enter the market at the higher price (area CEF).



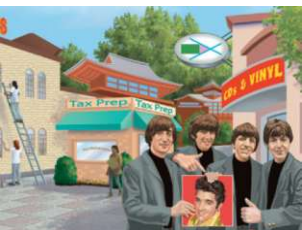
Market Efficiency

- Economic well-being of a society
 - Total surplus
 - Sum of consumer and producer surplus



Market Efficiency

- Total surplus = Consumer surplus + Producer surplus
 - Consumer surplus = Value to buyers – Amount paid by buyers
 - Producer surplus = Amount received by sellers – Cost to sellers
 - Amount paid by buyers = Amount received by sellers
- Total surplus = Value to buyers – Cost to sellers

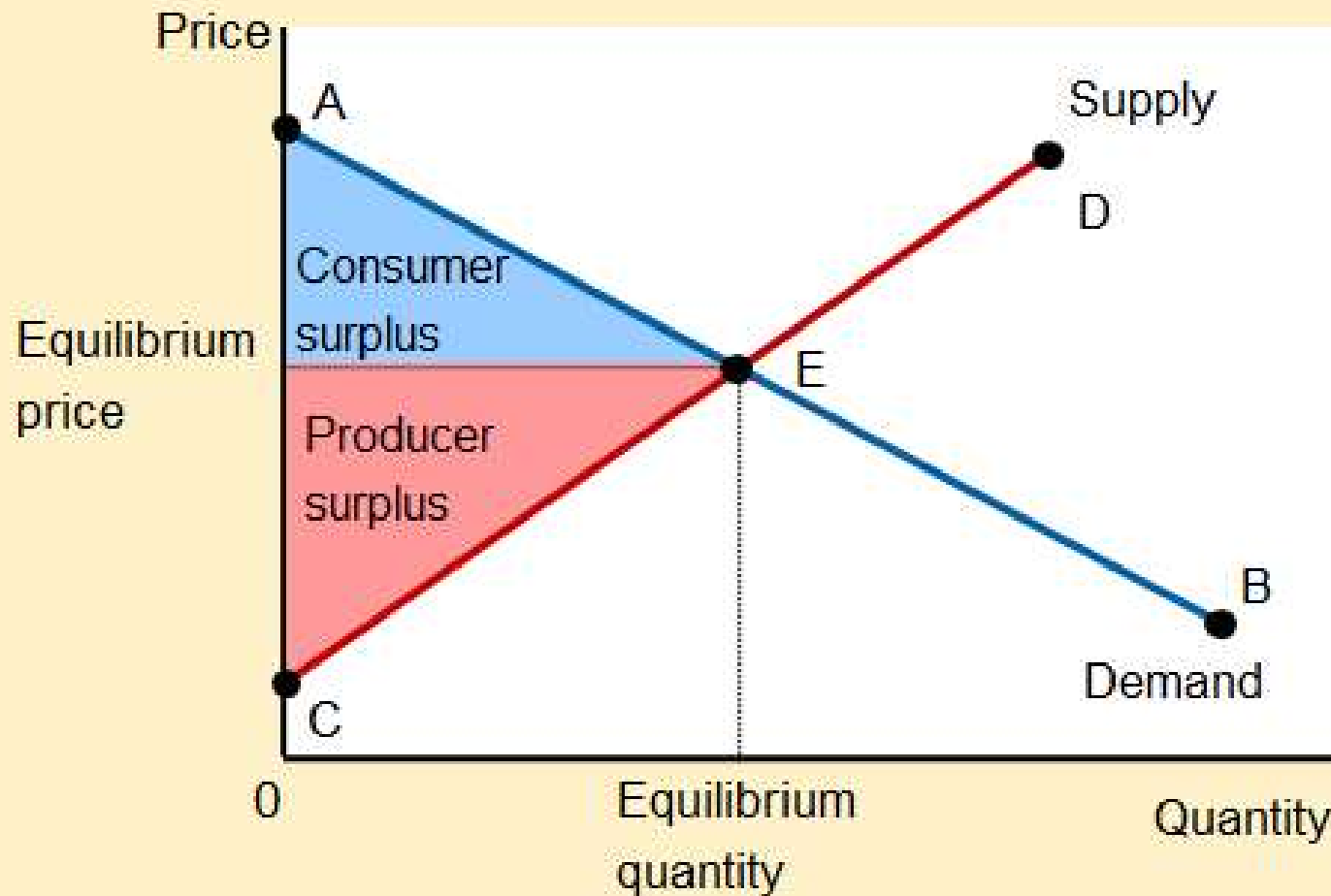


Market Efficiency

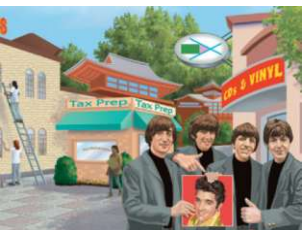
- Market outcomes

1. Free markets allocate the supply of goods to the buyers who value them most highly
 - Measured by their willingness to pay
2. Free markets allocate the demand for goods to the sellers who can produce them at the least cost

Figure 7 Consumer and Producer Surplus in the Market Equilibrium

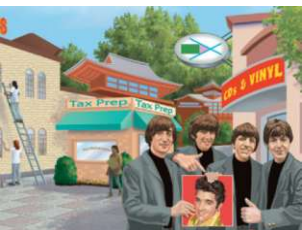


Total surplus—the sum of consumer and producer surplus—is the area between the supply and demand curves up to the equilibrium quantity



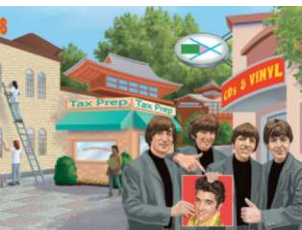
Market Efficiency, Part 6

- At market equilibrium, social planner
 - Cannot increase economic well-being by
 - Changing the allocation of consumption among buyers
 - Changing the allocation of production among sellers
 - Cannot rise total economic well-being by
 - Increasing or decreasing the quantity of the good



Market Efficiency, Part 7

- Market outcomes
 - Free markets produce the quantity of goods that maximizes the sum of consumer and producer surplus
- Market equilibrium
 - Efficient allocation of resources
- The benevolent social planner
 - “Laissez faire” = “let people do as they will”



Market Efficiency, Part 8

- Adam Smith's invisible hand
 - Takes all the information about buyers and sellers into account
 - Guides everyone in the market to the best outcome
 - Economic efficiency
- Free markets
 - Best way to organize economic activity